Reg. No. | | | | | | | | | |

Question Paper Code : 12639

B.E / B.TECH DEGREE EXAMINATION, NOV / DEC 2016

Third Semester

Common to B.E –Computer Science & Engineering and B.Tech-Information Technology

15MA304 – DISCRETE MATHEMATICS

(Regulations: Mepco – R2015)

Duration: 3 Hours Max. : 100 Marks

Answer ALL Questions

PART A – (10 × 2 = 20 Marks)

1. Find the power set of the set $\{0,1,2\}$.
2. Give an example for a function which is one-to-one but not onto.
3. What are the contrapositive and the inverse of the statement, "If it is raining, then it implies that the home team wins"?
4. Name the different proof techniques used to prove theorems.
5. Can a simple graph exist with 15 vertices each of degree 3? Reason out your answer.
6. Define, " An Euler path, a Hamiltonian path" of a graph G.
7. Define a group. Give an example of a monoid which is not a group.
8. When will an element in a ring is said to be a zero divisor?
9. Draw the Hasse diagram for $(\rho(A),\subseteq)$ where A = {a,b,c}.
10. State the principle of duality with respect to lattice.

PART B – (5 × 16 = 80 Marks)

11. a) i. Establish $\overline{A \cap B} = \bar{A} \cup \bar{B}$ analytically. (8 Marks)

11. a) ii. Find $f \circ g$ and $g \circ f$, where $f(x) = x^2 + 1$ and $g(x) = x + 2$, f and g are functions from R to R. Is $f \circ g$ equals $g \circ f$? (6 Marks)

11. a) iii. Represent the relation $\{(1,1),(1,2),(1,3),(2,2),(2,3),(3,3)\}$which is defined on a set $\{1,2,3\}$ with a relation matrix. (2 Marks)

OR

11. b) i. Let R be the relation on the set of ordered pairs of positive integers such that ((a,b),(c,d))∈R if and only if a+d=b+c. Show that R is an equivalence relation. (8 Marks)

11. b) ii. If $A = \left\{x \in R / x \neq \frac{1}{2}\right\}$ and $f: A \to R$ is defined by $f(x) = \frac{4x}{2x-1}$. Is this f invertible? If so, find a formula for f^{-1}. (6 Marks)

11. b) iii. Let A be the set of students who live within one mile of school and let B be the set of students who walk to classes. Describe the students in the set A∩B. (2 Marks)

12. a) i. Prove the following equivalence without using truth tables. $(\neg p \to \neg q) \to (q \to p) \equiv T$ (6 Marks)

12. a) ii. Show that the hypotheses, " It is not sunny this afternoon and it is colder than yesterday", "We will go swimming only if it is sunny", "If we do not go swimming, then we will take a local trip", and " If we take a local trip, then we will be home by sunset" lead to the conclusion " we will be home by sunset" using Rules of Inference for proposition. (10 Marks)

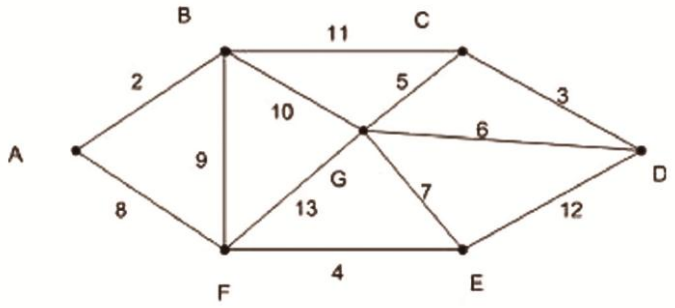
OR

12. b) i. Show that $(p \to r) \wedge (q \to r)$ and $(p \vee q) \to r$ are logically equivalent by developing truth table. (6 Marks)

12. b) ii. Show that the premises, " A student in this class has not read the book" and " Everyone in this class passed the first exam" imply the conclusion " Someone who passed the first exam has not read the book" using Rules of Inference for quantifiers. (10 Marks)

13. a) i. Prove that the number of vertices of odd degree in an undirected graph is even. (8 Marks)

13. a) ii. Find the minimum spanning tree for the weighted graph shown below by using Kruskal's algorithm.

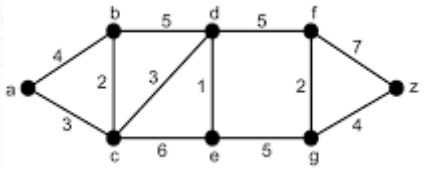


(8 Marks)

OR

13. b) i. Prove that a tree with n vertices has (n-1) edges. (8 Marks)

13. b) ii. Use Dijkstra's algorithm to find the shortest path between Vertices "a" and "z" in the weighted graph given below:



(8 Marks)

14. a) i. If $f:(G,*) \to (G',\Delta)$ is homomorphism with kernel K, then K is normal subgroup of G and the quotient group G/K is isomorphic to homomorphic image of G. (12 Marks)

14. a) ii. Define subgroup. Prove that intersection of any two subgroup is a subgroup. (4 Marks)

OR

14. b) i. State and prove Lagrange's theorem. (12 Marks)

14. b) ii. Define Cyclic group also prove that every cyclic group is abelian. (4 Marks)

15. a) i. State and prove distributive inequality in a lattice. (8 Marks)

15. a) ii. Define Boolean algebra. Is there a Boolean algebra with five elements? Justify. Also verify that $(D_{70}, /)$ is a Boolean algebra. (8 Marks)

OR

15. b) i. If $(L, \vee, \wedge)$ is a complemented distributive lattice, then for any $a, b \in L$, show that $\overline{(a \vee b)} = \bar{a} \wedge \bar{b}$ and $\overline{(a \wedge b)} = \bar{a} \vee \bar{b}$ (8 Marks)

15. b) ii. In a Boolean algebra, verify the following statements are equivalent

A) a+b= b B) a.b= a C) $a' + b = 1$ D) $a.b' = 0$ (8 Marks)